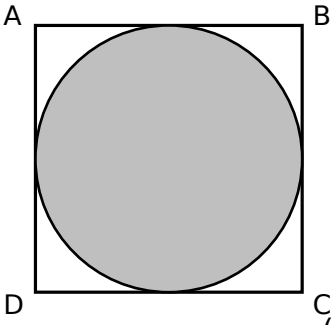
1. If $x^2 + y^2 = 25$ and $x = 3$, find the positive value of y.

(a) 2 (b) 4 (c) 5 (d) 6

2. In the figure below, ABCD is a square of side 4 cm and a circle is drawn inside it. Find the area of the shaded region.



(a) $16 - 4\pi$ cm²
(b) 4π cm²
(c) 16 cm²
(d) 8 cm²

3. Consider the following statements:

1. The Sun is a star.
2. The Moon is a planet.
3. Mercury is the planet closest to the Sun.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

4. Assertion (A): Ice floats on water.
Reason (R): Ice is less dense than liquid water.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Match List-I with List-II and select the correct answer.

List-I	**List-II**
A. Newton	1. Relativity
B. Einstein	2. Gravitation
C. Bohr	3. Atomic model
D. Darwin	4. Evolution

(a) A-2, B-1, C-3, D-4
(b) A-1, B-2, C-3, D-4
(c) A-2, B-1, C-4, D-3
(d) A-3, B-4, C-1, D-2

Directions (Q. 6-10): Read the following passage carefully and answer the questions that follow.

Honey bees live in large colonies that may contain tens of thousands of workers. A single queen lays the eggs, while the workers build the comb, collect nectar and pollen, and defend the hive. When a forager finds a rich source of food, she returns and performs a waggle dance. The angle of the dance tells the other bees the direction of the flowers relative to the sun, and the length of the waggle run tells them the distance. Scientists have shown that bees can learn colours, shapes and even simple patterns, which helps them to find the same flowers again.

6. Who lays the eggs in a honey bee colony?
   (a) The workers (b) The queen
   (c) The drones (d) The foragers

7. What does the angle of the waggle dance indicate?
   (a) Distance (b) Direction
   (c) Quantity (d) Colour

8. Which of these is NOT a task of the workers?
   (a) Building comb (b) Collecting nectar
   (c) Laying eggs (d) Defending the hive

9. The length of the waggle run tells the other bees the
   (a) direction (b) distance
   (c) colour (d) season

10. According to the passage, bees can learn
   (a) colours only (b) shapes only
   (c) colours, shapes and patterns (d) nothing

11. Which of the following figures has exactly one line of symmetry?



(a) (b) (c) (d)

12. A shopkeeper buys 120 notebooks at Rs 25 each and 80 pens at Rs 12 each. He sells three quarters of the notebooks at a profit of 20 percent and the remaining notebooks at a loss of 10 percent. He sells half of the pens at a profit of 25 percent and the other half at cost price. He also pays Rs 150 for transport and Rs 90 for packing material, which he treats as part of his cost. If he wants his overall profit on the whole transaction, including all expenses, to be exactly 15 percent, at what price must he sell a set of 10 extra notebooks that a school orders at the end of the month, given that the extra notebooks were bought at the same price as the others?

(a) Rs 412
(b) Rs 398
(c) Rs 420
(d) Rs 405

13. The formula of water is H_2O. What is the molar mass of CO_2?

(A) 44 g (B) 28 g (C) 32 g (D) 18 g

14. Find the value of 125 x 16. Write your answer as a whole number.

15. Which of the following numbers are prime?

(i) 2 (ii) 9 (iii) 11

(1) (i) only (2) (i) and (iii)
(3) (ii) only (4) all of these